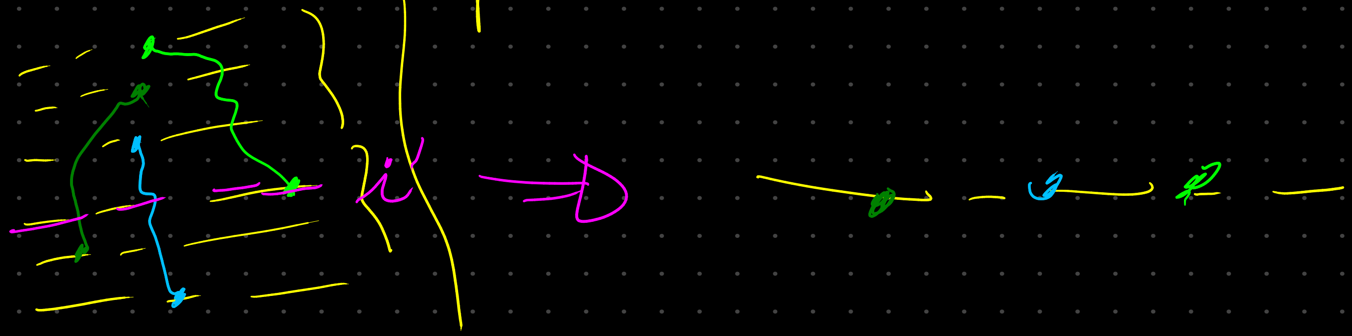
Recap before Meeting 26-05-2025

2SAT of G $\longrightarrow$ 2SAT of G^+ $\longrightarrow$ HT drawing of G^* $\longrightarrow$ X-monotone drawing of G^* $\longrightarrow$ level-planar drawing of G

$\forall i \in L(G)\ \exists i' \in L(G^*)\ \underline{\forall v \in V(G)}$ s.t. $\underline{l(v) = i} \Rightarrow$ its $e(v)$ crosses level i' in G^*

i' is part of the representation of i in G^*

we take the order in which the $e(v)$ $\forall v \in V(G)$ with $l(v) = i$ and determine the order of vs in G.



So we can draw a level planar drawing of G from a level planar drawing of G^*

we go line by line, the first level that we find all our stretch edge intersecting in it, we determine the order of the vertices corresponding to the mapped level in G.

(Γ^* to Γ ✓)

step 1 : $S(G) \longrightarrow S(G^+)$

$\forall i \in L(G)\ \forall v \in V(G)$ with $\ell(v) = i$

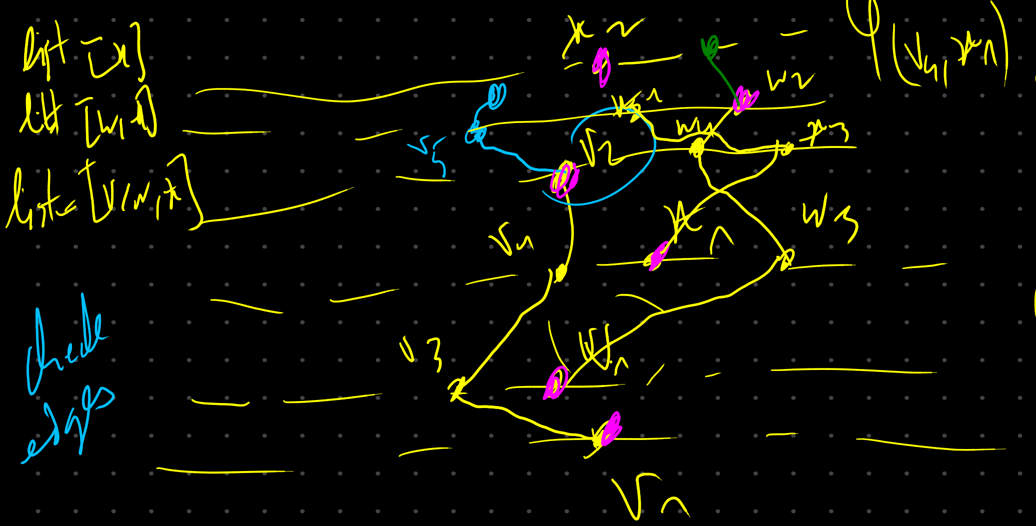
$\varphi^*(w_3, a_1) = \{(w_1 x)\}$

$\varphi^+(v_4, x_1)$

list $[\ell(i)]$

list $= [o(v_1), o(x_1)]$

$\varphi^+(x_1, v_3) = \{(w, v)\}$



$\ell(v_1) = 1$

$S(h) \to S(h^+) \to \Gamma^*$ of h^* (doesn't change any relative order)

Γ^* of h^* (H-T) $\to$ weak $H^t \to k$-monotone h^*

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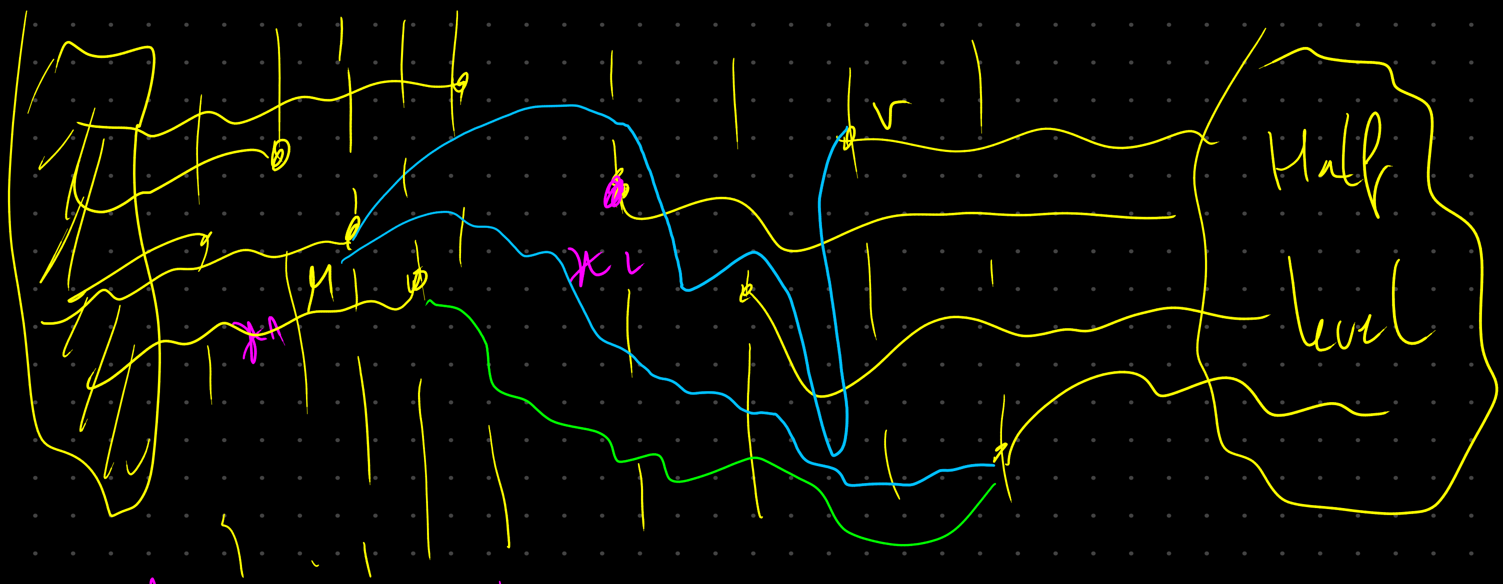
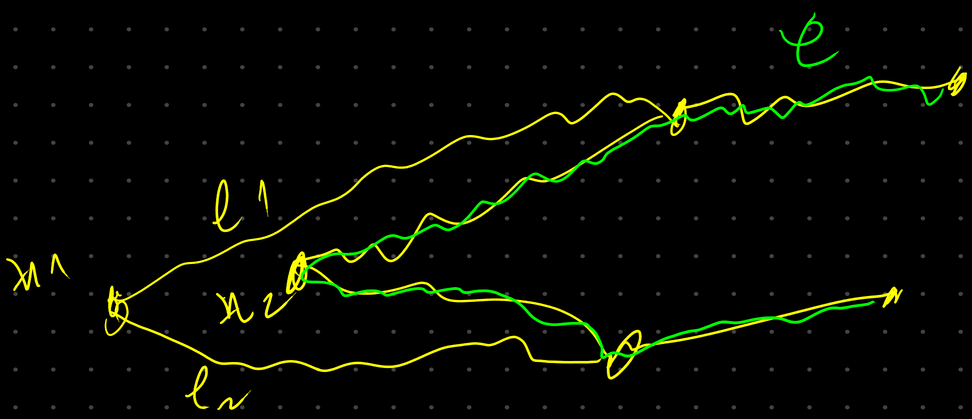
*) Γ^* is composed of L. components between levels corresponding to the same level of h. and then there is an edge (u, v) with $L_h(\ell(u)) \neq L_h(\ell(v))$, in this case we find ourselves in



case of h - m multiple comps.

the only time we can have this case, is when x_2 doesn't have any edge on the left. in this case we will always end up with multiple connected comps



$h - \{x_1\} \to 1$ c. comp

we can't have the case of $h - \{x_1\}$ multiple comps with

because the stretch edges are not connected to each other, so we can't have a H with

$e_1 > e_2$ & $f_1 < e_2$

$e_1 > f_2$ $f_1 < 1$

⇒ we are not in the case that use lemma

2.

3.